



Flexibility-based expansion planning of active distribution networks considering optimal operation of multi-community integrated energy systems

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ABSTRACT

Large-scale integration of distributed renewable energy into active distribution networks leads to a dramatic increase in flexibility requirements, and multi-community integrated energy systems may become important flexibility supplies owing to their multi-energy synergistic and complementary advantages. To fully utilize their flexible regulation potential, a flexibility-based hierarchical expansion planning model for active distribution networks considering optimal operation of multi-community integrated energy systems is proposed. In this model, the network and supply-demand flexibilities are integrated at the investment level to determine the installation types, locations, and capacities of distributed energy resources and power networks, and an interaction mechanism based on insufficient flexibility risk is designed at operating level to optimize operation strategies and balance flexibility and economy. Considering the renewable uncertainties, a model-free scenario generation method is introduced to capture the renewable seasonal and daily output characteristics without relying on complicated explicit feature modeling. Finally, a distributed iterative solution method is devised based on analytical target cascading theory. Simulations on a 54-node distribution system show the proposed approach utilizes the flexibility of multi-community integrated energy systems to reduce the configuration needs of active distribution networks for other flexible resources, improving the overall system economy and promoting renewable consumption.

1. Introduction

With the vigorous promotion of energy conservation and emissions reduction, active distribution networks (ADNs) that can support large-scale distributed renewable energy generators (REGs) have been rapidly developed [1]. However, owing to the strong volatility and uncertainty associated with REG outputs, ADNs must allocate sufficient controllable flexible resources to ensure efficient renewable consumption and reliable power supply [2]. Meanwhile, in the current context of vigorously developing multi-energy coupling and flexible inter-connecting technologies, an increasing number of community integrated energy systems (CIESs) are being integrated into ADNs owing to their multi-energy complementarity and ladder utilization abilities [3]. For ADNs, multi-community integrated energy systems (MCIESs) have bidirectional regulation effects on electric power, and their numerous internal controllable devices can function as flexible resources to cope

with REG volatility through electricity purchase and sale, thereby reducing the configuration requirements of ADNs for other flexible resources. However, MCIESs have their own energy supply methods, energy consumption characteristics, and stakeholders, their presence alters the composition structure of ADNs [4]. In this context, the expansion planning of restructured ADNs faces new requirements and challenges. Therefore, research on how to effectively utilize the flexible support potential of MCIESs for ADNs and realize synergistic planning of REGs and flexible resources is of considerable significance in postponing investments and promoting efficient renewable resource consumption.

1.1. Literature review

Several pioneering studies have investigated flexibility planning for ADNs. Reference [5] presented a joint planning model for distributed energy resources (DERs) and battery energy storage (BES) in ADNs to facilitate the consumption of renewable energy. Reference [6] proposed

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Nomenclature		
Abbreviations		
ADN	Active distribution network	
REG	Renewable energy generator	
CIES	Community integrated energy system	
MCIES	Multi-community integrated energy system	
DER	Distributed energy resources	
BES	Battery energy storage	
SOP	Soft open point	
GAN	Generative adversarial network	
WGAN-GP	Conditional Wasserstein generative adversarial network with gradient penalty	
ATC	Analytical target cascading	
Sets and Parameters		
Ω_R	Set of candidate installation REG	
Ω_F	Set of candidate installation flexible resources	
Λ_r / Λ_f	Sets of candidate installation nodes of type- r REG/type- f flexible resource	
Λ_S / Λ_{S0}	Sets of candidate installation nodes of newly built substations/expansion substations	
Λ_L	Set of candidate installation nodes of newly built lines	
Ω_M	Set of typical scenarios	
Ω_T	Set of scheduling periods	
Ω_D	Set of operating equipment in the AND	
Ω_N	Set of CIES in the ADN	
Ω_d	Set of operating equipment in CIES	
b	Discount rate	
c^r / c^f	Investment cost factors of type- r REG/type- f flexible resource	
c_i^S / c_i^{S0}	Construction and expansion costs of substations	
c^L / c^{TL}	Investment cost per unit length of newly built lines and supporting tie-lines for SOP	
G_{rated}^r	Unit installation capacity of type- r REG	
$z_{i,max}^r$	Maximum installation quantity of type- r REG at candidate installation node i	
∂	REG penetration rate	
β^{BES}	Maximum magnification factor of BES	
H_{gas}	Calorific value of natural gas	
$\phi_t^{ADN,u} / \phi_t^{ADN,d}$	Permissible upper limits of CVaR for the upward and downward insufficient flexibilities of ADN during period t	
Variables		
G_i^r / G_i^f	New installation capacities of type- r REG and type- f flexible resource at node i	
x_i^S / x_i^{S0}	Binary variable; it equals to 1 if installing newly built substation/expansion substation at node i	
x_{ij}^L / x_{ij}^{TL}	Binary variable; it equals to 1 if installing newly built line/supporting tie-lines for SOP on line ij	
$P_{i,t}^r / P_{i,t}^f$	Outputs of type- r REG and type- f flexible resource during period t	
$P_{n,t}^{CIES,buy}$	Electricity purchased by ADN from the n th CIES	
$P_{n,t}^{CIES,sell}$	Electricity sold by ADN to the n th CIES	
P_t^{grid}	Electricity purchased by ADN from main grid	
$I_{ij,t}$	Current of line ij during period t	
$U_{j,t}$	Voltage magnitude at node j during period t	
$P_{in,t}^i (H_{in,t}^i, C_{in,t}^i)$	Input powers of coupling devices in CIES	
$H_{dc,t}^{hes} / H_{ch,t}^{hes}$	Discharge and charge power of HES	

an ADN multistage distribution network planning method that considered active management of distributed generation units and demand response. In Ref. [7], a hierarchical multi-objective planning approach for ADNs is presented to improve the grid connection capabilities of photovoltaics (PVs) and wind turbines (WTs) by categorizing and managing the demand-side response. An ADN optimization and planning model with soft open point (SOP) was presented in Ref. [8] by considering the carbon emission cost. Reference [9] presented a dynamic expansion planning model that was integrated with seasonally transferred battery swapping stations and PVs. A coordination planning scheme of distributed generation, flexibility resource, and soft open point integrated with energy storage system was proposed in Ref. [10] to improve the flexible operation of active distribution networks. A flexible optimization planning model of distributed energy systems was proposed in Ref. [11] that ensured system flexibility by fully accommodating the PV outputs under multiple scenarios. Reference [12] presented a distribution network planning method that considered the flexibility of virtual power plants, and utilized the upward and downward flexible adjustment of distributed resources to ensure the complete consumption of PV outputs and load supply. Although these previous studies allocated flexible resources such as BES, demand response, and SOP to improve the flexibility of distribution networks, fewer have focused on the flexibility support ability of connected MCIESs, which have plentiful controllable devices and incur no additional investment cost during ADN planning. In addition, most studies treated flexibility requirement as rigid constraints in planning models and lacked an evaluation of insufficient flexibility risk, which may lead to overly conservative planning schemes.

The strong uncertainty of REG outputs significantly affects the stable operation and planning accuracy of distribution networks, making REG uncertainty modeling crucial for ADN planning. Scenario analysis is an

effective approach for describing uncertainty by transforming uncertain factors into multiple deterministic scenarios. Scholars have commonly used statistical methods such as time series simulations [13], scenario tree generation [14], and the Copula function for generating REG output scenarios [15]. Reference [16] utilized probabilistic statistical output models for WT and PV to portray their uncertainties. A multi-stage scenario tree of intraday uncertainty was established in Ref. [17] to describe the stochastic output of REGs. A wind power scenario generation method based on Copula functions and forecast errors was presented in Ref. [18]. However, REG outputs are affected by various complex factors such as geography, climate, and season. A primary challenge in statistical methods lies in the inability of simple models to precisely describe the impact of multiple factors, which makes deriving accurate model parameters difficult. With the rapid development of artificial intelligence, model-free scenario generation approaches are gaining traction in power systems owing to their ability to learn data distributions and generate new samples directly from historical data without making assumptions about the distribution that the data obeys [19]. The advantage over traditional statistical methods is the ability to mine the inherent distribution of uncertain variables, solve difficult modeling problems, and realize the unsupervised generation of scenarios. In Ref. [20], a conditional style-based generative adversarial network (GAN) was applied to generate REG scenarios. In Ref. [21], a GAN was leveraged to learn the spatio-temporal correlation of REG outputs, and the Wasserstein distance was introduced to enhance model training quality. Reference [22] proposed a federated learning-based framework for renewable scenario generation which achieved privacy protection.

Table 1 summarizes the main differences between the approach proposed in this study and the most relevant research in the field. Considering the above-mentioned previous studies, several research

Table 1

Comparison of the proposed approach with the related works.

Ref	Planning subject							REG and load uncertainties		Modeling method
	REG	MT	BES	SOP	Line	Substation	MCIES	Handling method	Insufficient flexibility risk	
[5]	✓	×	✓	×	×	×	×	Probability statistical method	×	Bi-level iterative programming
[6]	✓	×	×	×	✓	✓	×	×	×	Single-level centralized optimization
[7]	✓	×	×	×	✓	×	×	×	Insufficient power supply	Bi-level iterative programming
[8]	✓	×	×	✓	×	×	×	Probability statistical method	×	Bi-level coordinated optimization
[9]	×	×	✓	×	✓	×	×	×	×	Single-level centralized optimization
[11]	×	×	✓	×	×	×	×	Scenario tree	×	Tri-level optimization
[12]	×	×	✓	×	×	×	×	Frank-Copula function	REG curtailment rate	Bayesian distributed robust optimization
This paper	✓	✓	✓	✓	✓	✓	✓	Data-driven method: CWGAN-GP	CVaR of insufficient flexibility	Analytical target cascading-based distributed iterative optimization

gaps persist in flexibility planning for ADNs: (1) MCIESs have numerous and diversified flexible controllable resources, yet their flexibility regulation potential in ADN planning has not been fully explored; (2) The risk of insufficient flexibility is exacerbated with a high percentage of REGs, but previous studies have typically neglected to analyze or quantify this risk in-depth, instead adopted rigid constraints of flexibility requirements leading to overly conservative planning schemes or redundant flexible resources; (3) Most scenario generations are commonly modeled using statistical methods that require fitting specific model parameters. However, modeling becomes extremely difficult when diversity, correlation, and seasonality of renewable energy are considered simultaneously.

1.2. Contribution

To address the aforementioned gaps, this study presents a flexibility-based expansion planning framework for ADNs that effectively utilizes the operational flexibility of MCIESs in uncertain environments. Compared with the exiting research, the main contributions of this paper are as follows:

- (1) To fully utilize the flexibility of MCIESs, a flexibility-based ADN expansion planning method considering the optimal operation of MCIESs is proposed. This method can utilize the complementary coordination among multiple energy sources in MCIES operation to reduce the total planning cost, as well as improve the supply-demand and network flexibilities; Moreover, an ADN-MCIES interaction mechanism considering the risk of insufficient flexibility is developed to reduce the uncertainty risk of REG outputs and strike a balance between flexibility and economy.
- (2) To address the long- and short-term uncertainties of REG outputs, a model-free scenario generation approach based on the conditional Wasserstein generative adversarial network with gradient penalty (CWGAN-GP) is employed, which effectively captures the seasonal and daily output characteristics of REGs and avoid complex explicit feature modeling;
- (3) A distributed iterative solution method based on the analytical target cascading (ATC) theory is designed to simulate the game process among all parties and efficiently solve in parallel using only necessary data to be transmitted, which effectively protects the privacies of all parties and ensures convergence of the solution.
- (4) A case study is undertaken to validate the proposed methods and showcase the revenue potential of MCIES engaging in ADN planning using multi-energy coupling. The proposed method mitigates the penalty cost associated with insufficient flexibility and reduces the total cost encompasses investment and operation phases.

The remainder of this paper is structured as follows: Section 2 delineates the flexibility-based expansion planning framework of ADNs. The model-free scenario generation model based on CWGAN-GP is introduced in Section 3. Sections 4 and 5 detail the bi-level expansion planning model and its solution approach, respectively. Section 6 demonstrates the effectiveness of the proposed method through a case study. Finally, Section 7 provides a summary of the study.

2. Flexibility-based expansion planning framework

The flexibility of ADNs is reflected in the supply-demand balance of node flexible resources and the smoothness of network channels. The node flexible resources mainly include DERs such as microturbines (MTs), BESs, and substations. The network-side resources include the transmission lines and inter-nodal interconnections represented by SOPs. The proposed ADN hierarchical expansion planning framework considering the optimal operation of MCIESs is illustrated in Fig. 1, and consists of the following parts:

- (1) Considering the long-term seasonal variations in load demands, three typical seasonal (winter, summer, and transition) daily loads of ADN and MCIESs are selected. Next, considering the long- and short-term uncertainties of REGs, the CWGAN-GP scenario generation model is constructed to generate numerous REG output scenarios with seasonal and daily fluctuation characteristics of historical data, and scenario reduction is utilized to obtain typical scenario sets.
- (2) The ADN-MCIES coordinated operation level, which considers ADN-MCIES interaction mechanism based on insufficient flexibility risk is constructed to optimize their operation strategies by adopting the scenario-based stochastic optimization. The ADN and MCIESs iteratively operate with tie-line power as the transfer parameter until their interaction schemes are consistent. The structure and energy flow of a typical CIES investigated in this study are shown in Fig. 2.
- (3) The expansion planning model that integrates network and supply-demand flexibilities for ADNs is built, in which the installation types, locations, and capacities of DERs and power networks are optimized at the investment level by minimizing the annual investment cost and operating cost transmitted from the operation level. The operational management of DERs and power networks is carried out according to the established ADN-MCIES optimal scheduling model by adopting the installation strategy derived from the investment level.

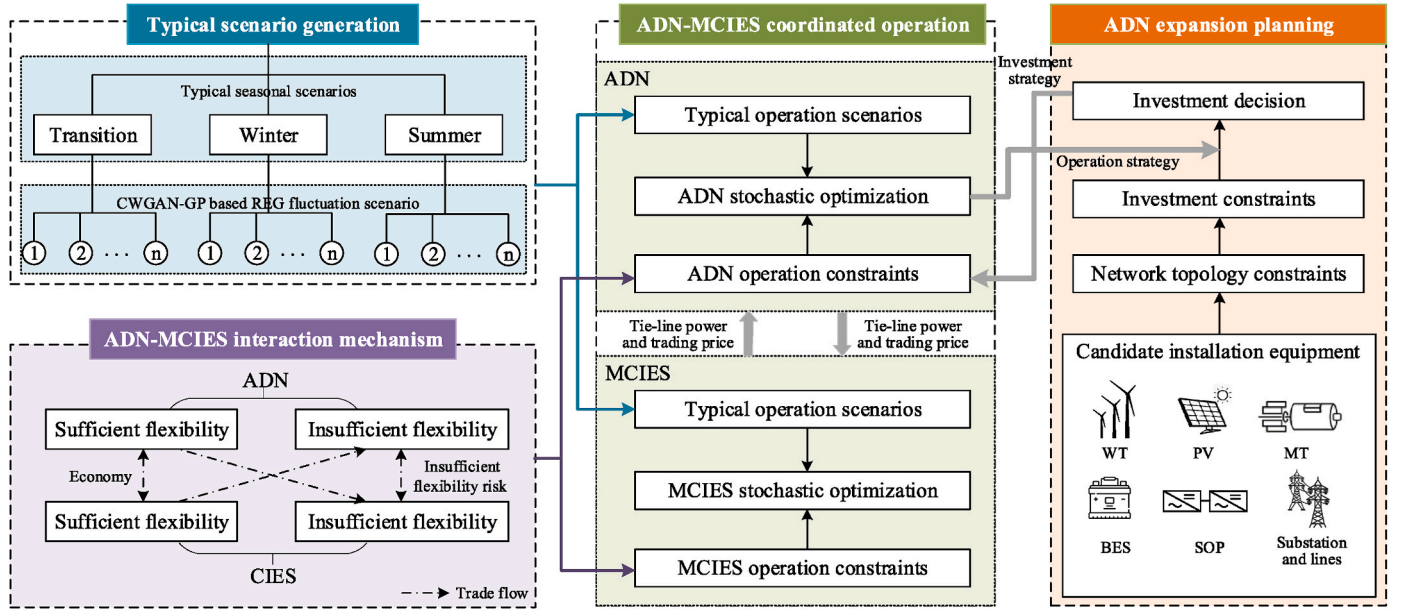


Fig. 1. ADN hierarchical expansion planning framework.

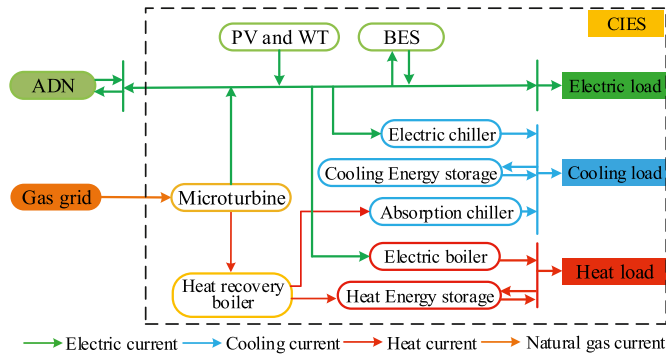


Fig. 2. Overall structure and energy flow of a typical CIES.

3. Conditional Wasserstein generative adversarial network-based scenario generation

The GAN, a deep learning model introduced by Goodfellow, comprises two key components: the generator G and the discriminator D [23]. The generator utilizes a black box pattern instead of a statistical model to avoid complex modeling of explicit features. Enhanced

authenticity of the generated samples is achieved through mutual game learning between these two networks. Conditional generative adversarial networks (CGANs) extend the traditional GAN by incorporating conditional information while retaining the game structure of the generator and discriminator. This augmentation empowers the generator to comprehend the mapping relationship of the sample probability distribution for specific conditions, leading to a substantial generalization effect on the specified type of data samples [24]. The network structure of the CGAN is depicted in Fig. 3.

As illustrated in Fig. 3, the noise, z , and condition, y , are combined as the inputs of the generator, and the outputs are the generated samples, x' . The discriminator processes real samples, x , conditions, and generated samples as inputs, yielding discrimination results. In this study, sample labels are utilized to generate renewable scenarios, considering various seasonal characteristics (i.e., winter, summer, and transition) as conditions.

The goal of the discriminator in the CGAN is to reduce the discriminative value of the generated scenarios, enhance that of the real scenarios, and ascertain whether the generated scenarios fulfill the specified conditions. Simultaneously, the generator strives to generate scenario samples that adhere to these conditions and elevate the discriminative value of the generated scenarios. Both networks are trained to achieve Nash equilibrium through a zero-sum game, enabling

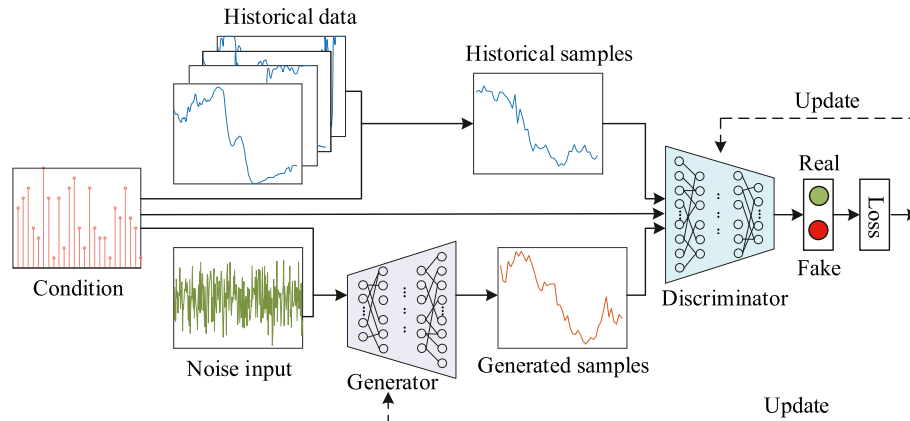


Fig. 3. Basic structure of CGAN.

the generator to effectively map corresponding conditional scenarios. Therefore, the objective function of the CGAN is [20]:

$$\min_G \max_D V(D, G) = E_{x \sim P(x)} [\log D(x|y)] + E_{x' \sim P(x')} [\log(1 - D(x'))] \quad (1)$$

where, P and E represent the probability distribution and its expected value; $D(\cdot)$ and $G(\cdot)$ are the outputs of the discriminator and generator.

To address the issues of instability and pattern collapse during CGAN training, the Wasserstein distance is utilized in the discriminator loss function to measure the probability distance between the real and generated scenarios [25].

$$\begin{aligned} \min C^{ADN} &= C_{inv}^{ADN} + C_{ope}^{ADN} \\ C_{inv}^{ADN} &= \sum_{r \in \Omega_R} \sum_{i \in \Lambda_r} \tau^r c^r G_i^r + \sum_{f \in \Omega_F} \sum_{i \in \Lambda_f} \tau^f c^f G_i^f + \sum_{i \in \Lambda_S} \tau^S c_i^S x_i^S + \sum_{i \in \Lambda_{S0}} \tau^{S0} c_i^{S0} x_i^{S0} + \sum_{ij \in \Lambda_L} \tau^L c^L x_{ij}^L L_{ij} + \sum_{ij \in \Lambda_{SOP}} \tau^{TL} c^{TL} x_{ij}^{TL} L_{ij} \\ C_{ope}^{ADN} &= 365 \sum_{m \in \Omega_M} p_{pro,m} C_{m,ope}^{ADN} \end{aligned} \quad (4)$$

$$W(P_x, P_{x'}) = \sup_{\|f_D\|_L \leq 1} E_{P(x)} D(x|y) - E_{P(x')} D(x'|y) \quad (2)$$

where, $\|f_D\|_L \leq 1$ indicates that the discriminator function needs to satisfy the 1-Lipschitz continuum.

In addition, a gradient penalty is introduced to make the discriminator approximate the 1-Lipschitz continuum, which results in the following objective function of CWGAN-GP [26]:

$$\begin{aligned} \min_G \max_D V(D, G) &= E_{x \sim P(x)} [\log D(x|y)] + E_{x' \sim P(x')} [\log(1 - D(x'))] \\ &\quad - \lambda E_{\hat{x} \sim P_{\hat{x}}} [(\|\nabla_{\hat{x}} D(\hat{x}|y)\| - 1)^2] \end{aligned} \quad (3)$$

where, λ is the penalty factor; $\hat{x} = \varepsilon x + (1 - \varepsilon)G(z)$, and ε obeys a uniform distribution over $[0, 1]$.

The training process of CWGAN-GP mainly consists of the following steps:

Step 1. Normalize the training samples and construct high-dimensional noises conforming to the standard normal distribution, splice the noises with the condition values corresponding to the training samples vertically and input them into the generator, which outputs the generated samples.

Step 2. Splice the condition values with the training and generated samples vertically, respectively, and then input them into the discriminator, which outputs the discriminative values for the real and generated samples.

Step 3. Calculate the loss functions of the generator and discriminator, and use the RMSprop optimization algorithm to update the weights of the generator and discriminator networks respectively. If the training is not finished, return to Step 1 for the next round of training.

Step 4. After training, extract the generator model from CWGAN-GP, input fixed condition values and n high-dimensional noises that conform to normal distribution, and the generator would output n renewable generation output scenarios that meet the condition values.

Upon generating numerous renewable scenarios for three seasons using CWGAN-GP, typical scenarios are obtained using K-Medoids and then applied to the operation level of the hierarchical planning model [27].

4. Hierarchical expansion planning model

Based on the flexibility expansion planning framework proposed in Section 2, the ADN flexibility expansion planning model considering the optimal operation of MCIESs is constructed.

4.1. Investment level

The minimum total cost composed of the annualized investment cost C_{inv}^{ADN} and annualized operating cost C_{ope}^{ADN} is adopted as the objective function to be optimized as follows:

where, $\tau = b(1 + b)^\sigma / [(1 + b)^\sigma - 1]$ is the coefficient for converting the present value of equipment to its annual value, in which b and σ are the discount rate and lifespan, respectively; Ω_R is the set of candidate installation REGs, including PV and WT; Ω_F is the set of flexible resources including MT, BES, and SOP; Λ_r , Λ_f , Λ_S , Λ_{S0} , Λ_L , and Λ_{SOP} are sets of candidate installation nodes for REGs, flexible resources, newly built substations, expansion substations, newly built lines, and supporting tie-lines for SOP, respectively; c^r and c^f are the investment cost factors of REG and flexible resources; c_i^S and c_i^{S0} are the construction and expansion costs of substations; c^L and c^{TL} are the investment cost per unit length of newly built lines and supporting tie-lines for SOP; G_i^r and G_i^f are the new installation capacities of type- r REG and type- f flexible resources at node i , respectively; x_i^S and x_i^{S0} are 0–1 decision variables for whether newly built substations and expansion substations are installed at node i ; x_{ij}^L and x_{ij}^{TL} are 0–1 decision variables for whether newly built lines and supporting tie-lines for SOP are installed on line ij ; L_{ij} is the length of line ij ; Ω_M denotes the set of typical scenarios; $p_{pro,m}$ and $C_{m,ope}^{ADN}$ are the probability and daily operating cost of scenario m , respectively.

The investment constraints include:

(1) REG installation capacity and quantity constraints

$$\begin{aligned} G_i^r &= z_i^r G_{rated}^r \\ 0 &\leq z_i^r \leq z_{i,max}^r \\ \sum_{r \in \Omega_R} \sum_{i \in \Lambda_r} G_i^r &\leq \partial P_{max}^L \end{aligned} \quad (5)$$

where, G_{rated}^r is the unit installation capacity of type- r REG; $z_{i,max}^r$ is the maximum installation quantity of type- r REG at candidate installation node i ; ∂ is the REG penetration rate; P_{max}^L is the maximum load demand.

(2) Flexible resource installation capacity and quantity constraints

The installation capacity and rated power constraints of BES are as follows:

$$\begin{aligned} 0 &\leq G_i^{BES} \leq G_{i,max}^{BES} \\ 0 &\leq S_i^{BES} \leq S_{i,max}^{BES} \\ G_i^{BES} &\leq S_i^{BES} \rho^{BES} \end{aligned} \quad (6)$$

where, $G_{i,\max}^{BES}$ and $S_{i,\max}^{BES}$ are the maximum installation capacity and rated power of BES at node i , respectively; ρ^{BES} is the maximum magnification factor.

The installation capacity constraint of MT is given as:

$$0 \leq G_i^{MT} \leq G_{i,\max}^{MT} \quad (7)$$

where, $G_{i,\max}^{MT}$ is the maximum installation capacity of MT at node i .

The installation capacity and quantity constraints of SOP are [28]:

$$\begin{aligned} G_{ij}^{SOP} &= z_{ij}^{SOP} G_{rated}^{SOP} \\ 0 &\leq z_{ij}^{SOP} \leq z_{ij,\max}^{SOP} x_{ij}^{TL} \end{aligned} \quad (8)$$

where, G_{rated}^{SOP} is the unit installation capacity of SOP; $z_{ij,\max}^{SOP}$ is the maximum installation quantity of SOP on line ij .

(3) Network topology constraints

Given that the candidate installation locations for SOPs are at the interconnection switches, coordinated planning necessitates the establishment of a radiation network and tie-lines. To ensure that the tie-lines constructed for SOP installation do not intersect with the selected lines of the radial network, a pertinent logical constraint is introduced, as follows:

$$x_{ij}^{TL} + x_{lj}^L \leq 1 \quad (9)$$

A restructured ADN needs to maintain a radial and connected grid structure without loops or islands. The following equations are then used for constraints [29]:

$$\begin{aligned} W_{ij} + W_{ji} &= \delta_{ij} \quad i, j = 1, \dots, q \\ \sum_{j \in \Omega_L(i)} W_{ij} &= 1 \quad i = 2, \dots, q \\ W_{ij} &= 0 \quad j \in \Omega_L(1) \end{aligned} \quad (10)$$

where, δ_{ij} is the 0–1 variable that describes the on-off state of branch ij , with 0 being disconnected and 1 being closed; q is the number of ADN nodes; W_{ij} is a 0–1 variable, where 1 indicates that the current of branch ij flows from node j to node i , and 0 indicates the opposite; $\Omega_L(i)$ is the set of nodes connected to node i .

4.2. Operating level

$$\begin{aligned} \min C_{ope}^{ADN} &= C_1 + C_2 + C_3 + C_4 + C_5 \\ C_1 &= \sum_{t \in \Omega_T} \left[\gamma_{gas}^{ADN} (P_t^{MT} / \eta^{MT} / H_{gas}) \right] \Delta t \\ C_2 &= \sum_{i \in \Lambda_S} \kappa_i x_i^S + \sum_{i \in \Lambda_{S0}} \kappa_i x_i^{S0} + \sum_{ij \in \Lambda_L} \kappa_{ij} x_{ij}^L + \sum_{ij \in \Lambda_{SOP}} \kappa_{ij} x_{ij}^{TL} + \sum_{t \in \Omega_T} \left(\sum_{r \in \Omega_R} \kappa_r P_{r,t}^r + \sum_{f \in \Omega_F} \kappa_f P_{f,t}^f \right) \Delta t \\ C_3 &= \sum_{t \in \Omega_T} \sum_{n \in \Omega_N} \left(\gamma_t^{buy} P_{n,t}^{CIES,buy} - \gamma_t^{sell} P_{n,t}^{CIES,sell} \right) \Delta t \\ C_4 &= \sum_{t \in \Omega_T} \gamma_t^{grid} P_t^{grid} \Delta t \\ C_5 &= \sum_{t \in \Omega_T} \left[\sum_{r \in \Omega_R} \sum_{i \in \Lambda_r} \left(P_{i,t}^{r,E} - P_{i,t}^{r,E} \right) \gamma^{CR} + P_t^{CL} \gamma^{CL} \right] \Delta t \end{aligned} \quad (12)$$

An ADN-MCIES flexibility interaction mechanism and coordinated optimization scheduling models of ADN and MCIESs are constructed, so that MCIESs can provide upward or downward flexibility adjustment through the coupling and complementarity among multiple energy

sources for ADN when its upward or downward flexibility is insufficient.

4.2.1. Flexibility interaction mechanism

Owing to the volatility and uncertainty of REGs and load demand, insufficient flexibility risk exists in each CIES and ADN. Scientific assessment of this risk is useful for assisting decision-makers in choosing a rational planning scheme. Therefore, based on the conditional value at risk (CVaR) theory in economics [30], a CVaR of insufficient flexibility is proposed to quantify the risk and discretized by combining typical scenarios as follows:

$$\begin{aligned} F_{ren,t}^{CVaR} &= \alpha_1 + \frac{1}{1-\beta} \sum_{m \in \Omega_M} p_{pro,m} P_t^{CR} \\ F_{load,t}^{CVaR} &= \alpha_2 + \frac{1}{1-\beta} \sum_{m \in \Omega_M} p_{pro,m} P_t^{CL} \end{aligned} \quad (11)$$

where, $F_{ren,t}^{CVaR}$ and $F_{load,t}^{CVaR}$ are the CVaRs of renewable curtailment and load shedding, respectively, during period t (i.e., the expected value that exceeds the flexibility deficit threshold at a given confidence level β); α_1 and α_2 are the boundary values of renewable curtailment and load shedding; P_t^{CR} and P_t^{CL} are the powers of renewable curtailment and load shedding, which reflects the system supply-demand flexibility.

Based on the CVaR of insufficient flexibility, the following ADN-MCIES flexibility interaction mechanism is designed:

- (1) Flexible resources are supplied by CIES to ADN when the CIES flexibility is sufficient and the ADN flexibility is insufficient;
- (2) Flexible resources are provided by ADN to CIES when the ADN flexibility is sufficient and the CIES flexibility is insufficient;
- (3) The CIES and ADN perform flexibility interaction based on economics when both flexibilities are sufficient;
- (4) The CIES and ADN perform flexibility interaction based on the CVaR of insufficient flexibility when both flexibilities are insufficient.

4.2.2. Optimal scheduling of active distribution network

The ADN operating level manages the operation of the DERs and power network with the optimization goal of minimizing the operating cost for each typical day. The scenario m subscript is omitted owing to the consistent scheduling model for each typical scenario.

where, C_1 , C_2 , C_3 , C_4 , and C_5 are the fuel cost of MT, operation and maintenance (O&M) cost of substations, lines, REGs, and flexible resources, interaction cost with MCIESs, power purchase cost from the main grid, and insufficient flexibility penalty cost, respectively; γ_{gas}^{ADN} is

the price of natural gas, and H_{gas} is the calorific value of natural gas; P_t^{MT} is the MT output during period t ; η^{MT} is the electrical efficiency of MT; κ_i is the O&M cost factor of equipment i ; $P_{i,t}^r$ and $P_{i,t}^f$ are the outputs of REGs and flexible resources during period t ; Ω_N is the set of CIESs in the ADN; γ_t^{buy} and $P_{n,t}^{CIES,buy}$ are the price and power of electricity purchased by the ADN from the n th CIES, respectively; γ_t^{sell} and $P_{n,t}^{CIES,sell}$ are the price and electricity sold by the ADN to the n th CIES, respectively; γ_t^{grid} and $P_{i,t}^{grid}$ are the price and electricity purchased by the ADN from the main grid; $P_{i,t}^{F,E}$ and $P_{i,t}^E$ are the expected output and consumption of REGs during period t ; γ^{CR} and γ^{CL} are the penalty cost factors of renewable curtailment and load shedding; Δt is the scheduling interval; and Ω_T is the set of scheduling periods.

The constraints for optimal scheduling of the ADN are described below.

(1) AC power flow constraints

$$\begin{aligned} \sum_{k \in \Omega_L(j,:)} P_{jk,t} - \sum_{i \in \Omega_L(:,j)} (P_{ij,t} - I_{ij,t}^2 R_{ij}) &= \sum_{d \in \Omega_D} P_{j,t}^d - (P_{j,t}^L - P_{j,t}^{CL}) + \sum_{n \in \Omega_N} (P_{n,t}^{CIES,buy} \\ &- P_{n,t}^{CIES,sell}) \sum_{k \in \Omega_L(j,:)} Q_{jk,t} - \sum_{i \in \Omega_L(:,j)} (Q_{ij,t} - I_{ij,t}^2 X_{ij}) \\ &= Q_{j,t}^L U_{j,t}^2 = U_{j,t}^2 - 2(R_{ij} P_{ij,t} + X_{ij} Q_{ij,t}) + (R_{ij}^2 + X_{ij}^2) I_{ij,t}^2 = \frac{P_{ij,t}^2 + Q_{ij,t}^2}{U_{i,t}^2} \end{aligned} \quad (13)$$

where, $\Omega_L(j,:)$ is the set of end nodes of branches with j as the first node, and $\Omega_L(:,j)$ is the set of first nodes of branches with j as the end node; Ω_D is the set of operating equipment in the ADN, including REGs, MTs, BESs, SOPs, and substations; $P_{jk,t}$, $P_{ij,t}$ and $Q_{jk,t}$, $Q_{ij,t}$ are the active and reactive powers of lines jk and ij during period t ; R_{ij} and X_{ij} are the resistance and reactance of line ij ; $P_{j,t}^d$ is the active power injected into node j during period t by operating equipment d ; $P_{j,t}^L$ and $Q_{j,t}^L$ are the original active and reactive loads at node j ; $I_{ij,t}$ is the current of line ij ; and $U_{j,t}$ is the voltage magnitude at node j during period t .

(2) Security constraints

$$\begin{aligned} U_{i,\min}^2 &\leq U_{i,t}^2 \leq U_{i,\max}^2 \\ 0 &\leq I_{ij,t}^2 \leq I_{ij,\max}^2 \end{aligned} \quad (14)$$

where, $U_{i,\min}$ and $U_{i,\max}$ are the minimum and maximum values of the node voltage, respectively; $I_{ij,\max}$ is the maximum current of line ij .

(3) Substation power constraints

$$\begin{aligned} \sqrt{(P_{i,t}^S)^2 + (Q_{i,t}^S)^2} &\leq x_i^S G_{i,rated}^S, \forall i \in \Omega_S \\ \sqrt{(P_{i,t}^{S0})^2 + (Q_{i,t}^{S0})^2} &\leq G_{i,rated}^{S0} + x_i^{S0} G_{i,rated}^{S0,new}, \forall i \in \Omega_{S0} \end{aligned} \quad (15)$$

$$\min C_{ope}^{CIES} = \sum_{t \in \Omega_T} \left[\gamma_{gas} (P_t^{mt} / \eta^{mt} / H_{gas}) + \sum_{i \in \Omega_d} P_t^i \kappa^i + \gamma^{dr} \times \min(P_t^{tse}, 0) + (\gamma^{cr} P_t^{cr} + \gamma^{cl} P_t^{cl}) + (\gamma_t^{sell} P_t^{ADN,buy} - \gamma_t^{buy} P_t^{ADN,sell}) \right] \Delta t \quad (19)$$

where, $P_{i,t}^S$, $Q_{i,t}^S$ and $P_{i,t}^{S0}$, $Q_{i,t}^{S0}$ are the active and reactive power outputs of newly built and expansion substations; $G_{i,rated}^S$ is the rated capacity of newly built substations; $G_{i,rated}^{S0}$ is the rated capacity of existing substations; $G_{i,rated}^{S0,new}$ is the rated expansion capacity of existing substations.

(4) SOP power constraints

SOPs are generally back-to-back voltage source converters consisting of high-power, fully-controlled power electronic components [31]. The steady state control mode of $PQ-V_{dc}Q$ is employed in this study, i.e., one converter realizes stable control of the DC voltage and the other converter realizes flexible control of the transmitted power [32]. Although SOP has certain losses, it can be ignored relative to the entire distribution system. Therefore, the power output constraints of SOPs are:

$$\begin{aligned} P_{i,t}^{SOP} + P_{j,t}^{SOP} &= 0 \\ \sqrt{(P_{i,t}^{SOP})^2 + (Q_{i,t}^{SOP})^2} &\leq P_{ij,\max}^{SOP} \\ \sqrt{(P_{j,t}^{SOP})^2 + (Q_{j,t}^{SOP})^2} &\leq P_{ij,\max}^{SOP} \end{aligned} \quad (16)$$

where, $P_{i,t}^{SOP}$, $Q_{i,t}^{SOP}$ and $P_{j,t}^{SOP}$, $Q_{j,t}^{SOP}$ are the active and reactive outputs of SOP at nodes i and j , respectively; $P_{ij,\max}^{SOP}$ is the maximum active power of SOP on line ij .

(5) Flexibility interaction power constraints with MCIES

$$\begin{aligned} 0 &\leq P_{n,t}^{CIES,buy} \leq \delta_{n,t}^{CIES,buy} P_{n,\max}^{CIES,buy} \\ 0 &\leq P_{n,t}^{CIES,sell} \leq \delta_{n,t}^{CIES,sell} P_{n,\max}^{CIES,sell} \\ \delta_{n,t}^{CIES,buy} + \delta_{n,t}^{CIES,sell} &\leq 1 \end{aligned} \quad (17)$$

where, $\delta_{n,t}^{CIES,buy}$ and $\delta_{n,t}^{CIES,sell}$ are 0–1 variables describing the purchasing and selling status of the ADN from the n th CIES, respectively; $P_{n,\max}^{CIES,buy}$ and $P_{n,\max}^{CIES,sell}$ are the maximum power purchased and sold by the ADN from the n th CIES during period t , respectively.

(6) Insufficient flexibility risk constraints

$$\begin{aligned} F_{ren,t}^{ADN,CVaR} &\leq \phi_t^{ADN,u} \\ F_{load,t}^{ADN,CVaR} &\leq \phi_t^{ADN,d} \end{aligned} \quad (18)$$

where, $\phi_t^{ADN,u}$ and $\phi_t^{ADN,d}$ are the permissible upper limits of the CVaR for the upward and downward insufficient flexibilities of ADN during period t .

In addition, the MT and BES operating constraints as well as the REG consumption and load shedding constraints are commonly used and thus not discussed in this paper; the details can be found in our previous work [21].

4.2.3. Optimal scheduling of multi-community integrated energy system

The MCIES operating level optimizes the operation strategies of REGs and controllable devices in each CIES by minimizing the operating cost for each typical day. Similar to the ADN scheduling model, the scenario m subscript is omitted.

where, γ_{gas} is the price of natural gas purchased by CIES; P_t^{mt} and η^{mt} are the output and electrical efficiency of MT in CIES; Ω_d is the set of operating equipment in CIES, including REG, MT, heat recovery boiler (HRB), electric chiller (EC), absorption chiller (AC), electric boiler (EB), BES, heat energy storage (HES), and cooling energy storage (CES); P_t^i is

the output of operating equipment i during period t ; γ^{dr} is the demand response compensation cost factor, and P_t^{se} is the time-shiftable electric load during period t ; P_t^{cr} and P_t^{cl} are the renewable curtailment and load shedding powers in CIES during period t ; $P_t^{ADN,buy}$ and $P_t^{ADN,sell}$ are the electricity purchased and sold by CIES from the ADN, respectively.

The constraints for CIES optimal scheduling include:

(1) Power balance constraints

$$\begin{cases} P_t^{wt} + P_t^{pv} + P_t^{mt} + P_{dc,t}^{bes} + P_t^{ADN,buy} = P_t^l + P_t^{se} + P_{ch,t}^{bes} + P_{in,t}^{eb} + P_{in,t}^{ec} - P_t^{cl} + P_t^{ADN,sell} \\ H_{out,t}^{rb} + H_{out,t}^{eb} + H_{dc,t}^{hes} = H_t^l + H_{ch,t}^{hes} + H_{in,t}^{ac} \\ C_{out,t}^{ac} + C_{out,t}^{ec} + C_{dc,t}^{ces} = C_t^l + C_{ch,t}^{ces} \end{cases} \quad (20)$$

where, P_t^{wt} and P_t^{pv} are the consumption power of WT and PV during period t ; P_t^l , H_t^l , and C_t^l are the electric, heat, and cooling loads, respectively; $P_{in,t}^i (H_{in,t}^i, C_{in,t}^i)$ and $P_{out,t}^i (H_{out,t}^i, C_{out,t}^i)$ are the input and output powers of coupling devices in CIES; $P_{dc,t}^{bes}$ ($P_{ch,t}^{bes}$), $H_{dc,t}^{hes}$ ($H_{ch,t}^{hes}$), and $C_{dc,t}^{ces}$ ($C_{ch,t}^{ces}$) are the discharge and charge powers of BES, HES, and CES, respectively.

(2) Flexibility interaction power constraints with ADN

$$\begin{aligned} 0 &\leq P_t^{ADN,buy} \leq \delta_t^{ADN,buy} P_{max}^{ADN,buy} \\ 0 &\leq P_t^{ADN,sell} \leq \delta_t^{ADN,sell} P_{max}^{ADN,sell} \\ \delta_t^{ADN,buy} + \delta_t^{ADN,sell} &\leq 1 \end{aligned} \quad (21)$$

where, $\delta_t^{ADN,buy}$ and $\delta_t^{ADN,sell}$ are 0–1 variables describing the purchasing and selling status of the CIES from the ADN; $P_{max}^{ADN,buy}$ and $P_{max}^{ADN,sell}$ are the maximum power purchased and sold by CIES from the ADN during period t , respectively.

(3) Insufficient flexibility risk constraints

$$\begin{aligned} F_{ren,t}^{CIES,CVaR} &\leq \phi_t^{CIES,u} \\ F_{load,t}^{CIES,CVaR} &\leq \phi_t^{CIES,d} \end{aligned} \quad (22)$$

where, $\phi_t^{CIES,u}$ and $\phi_t^{CIES,d}$ are the permissible upper limits of the CVaR for the upward and downward insufficient flexibilities of CIES during period t , respectively.

Other operating constraints in each CIES, including the power output and ramp rate constraints of MT and multi-energy coupling equipment [33], capacity constraints of BES, HES, and CES [34], as well as the REG consumption and load shedding constraints [35], are commonly used, and are thus not discussed in this paper.

5. Model solving

The ADN hierarchical expansion planning model constructed above contains interactive iterations between the investment and operating levels, as well as between the ADN and MCIES, which considerably increase the solving complexity. Therefore, the following model

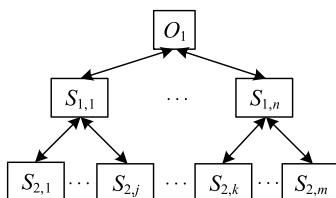


Fig. 4. Typical ATC system structure.

conversion and iterative algorithm are presented to facilitate the solving process.

5.1. Hierarchical model conversion

Considering that ADN and MCIES belong to different stakeholders, the interactive iteration between ADN and MCIES is retained, and the investment and operating levels of ADN are converted into a single-level model through variable replacement to avoid complicated parameter passing.

In terms of REGs, the parameter $P_{i,t}^{r,E}$ in Eq. (12) is replaced using the following equation:

$$\begin{aligned} P_{i,t}^{WT,E} &= G_t^{WT} P_{unit,i,t}^{WT,E} \\ P_{i,t}^{PV,E} &= G_i^{PV} P_{unit,i,t}^{PV,E} \end{aligned} \quad (23)$$

where, $P_{unit,i,t}^{WT,E}$ and $P_{unit,i,t}^{PV,E}$ are the per unit expected outputs of WT and PV.

For flexible resources, the rated power of MT, BES, and SOP in the ADN operational constraints are replaced with:

$$\begin{aligned} P_{i,max}^{MT} &= G_i^{MT} \\ P_{i,max}^{BES} &= G_i^{BES} \\ P_{ij,max}^{BES} &= G_{ij}^{SOP} \end{aligned} \quad (24)$$

5.2. Analytical target cascading-based distributed solving

The iterations between ADN and MCIES are addressed by leveraging the ATC algorithm. ATC, a recent mathematical optimization method known for its superior convergence, couples systems through decision variables in hierarchical structures [36]. A typical ATC structure is shown in Fig. 4, in which O_1 is the main system, $S_{1,1}$ – $S_{1,n}$ are sub-generation systems, $S_{2,1}$ – $S_{2,j}$ and $S_{2,k}$ – $S_{2,m}$ are second-generation systems. Each system can set its own optimization model, where the variables contained in the optimization model of each level that are common with other levels are coupled variables. The information flow is transmitted level by level from top to bottom, where the lower level system optimizes its own performance by moving closer to the coupled variables passed from the upper level system while satisfying self-constraints. The optimized coupled variables of the lower level system are then fed back as responses to the upper level system, which receives them as constants and adds them to its objective for optimization, similar to the lower level systems. This loop continues until the conditions are satisfied.

In the proposed model, the ADN and MCIES are coupled through tie-line powers. Following the initial optimal dispatch of the ADN, the optimized tie-line power, $\bar{P}_{n,t}^{ADN}$, is passed on as a known parameter to the MCIES. Subsequently, a penalty function is added to the objective function of each CIES to reduce the deviation of the interaction power from $\bar{P}_{n,t}^{ADN}$ as follows:

$$\min C_{n,ope}^{CIES} + \sum_{t=1}^T \omega_{n,t} |P_{n,t}^{ADN,buy} - P_{n,t}^{ADN,sell} - \bar{P}_{n,t}^{ADN}| \quad (26)$$

where, $\omega_{n,t}$ is the penalty function multiplier of CIES n during period t .

Similarly, after the optimization of each CIES is completed, the tie-line power, $\bar{P}_{n,t}^{CIES}$, is transmitted to the ADN. A penalty function is also added to the objective function of the ADN to reduce the deviation of its interaction power from $\bar{P}_{n,t}^{CIES}$:

$$\min C^{ADN} + \sum_{t=1}^T \sum_{n=1}^N \omega_{n,t} |P_{n,t}^{CIES,buy} - P_{n,t}^{CIES,sell} - \bar{P}_{n,t}^{CIES}| \quad (27)$$

This process is repeated until the following convergence criteria are satisfied:

Table 2

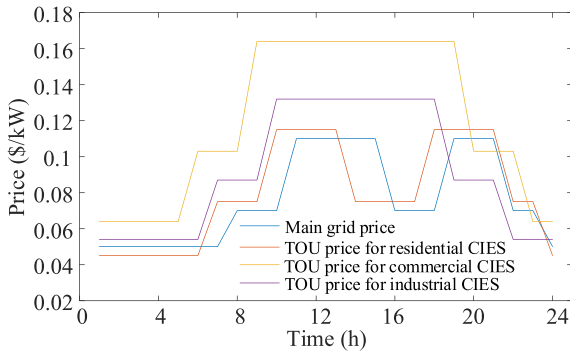
ADN parameters.

Parameter	Value	Parameter	Value	Parameter	Value
b	0.08	H_{gas} (kWh/m ³)	9.7	η^{MT}	0.44
σ (year)	20	z_{max}^*	50	κ^{WT} (\$/kW)	0.0064
c^{WT} (\$/kW)	814	∂	0.7	κ^{PV} (\$/kW)	0.0044
c^{PV} (\$/kW)	525	$G_{\text{max}}^{\text{BES}}$ (kWh)	2000	κ^{BES} (\$/kW)	0.01
c^{MT} (\$/kW)	202	$S_{\text{max}}^{\text{BES}}$ (kW)	500	κ^{MT} (\$/kW)	0.0058
c^{BES} (\$/kW)	218	β^{BES}	0.25	κ^L (\$)	436
c^{SOP} (\$/kW)	145	G_{max}^{MT} (kW)	10000	γ^{CR} (\$/kW)	0.1
c^L (\$/km)	35650	G_{rated}^* (kW)	100	γ^{CL} (\$/kW)	0.2
c^{TL} (\$/km)	35650	$G_{\text{rated}}^{\text{SOP}}$ (kW)	100	$\phi^{ADN,u}$ (kW)	500
γ_{GAS} (\$/m ³)	0.4	$s_{\text{max}}^{\text{SOP}}$	50	β	0.95

Table 3

MCIES parameters.

Parameter	Residential CIES	Commercial CIES	Industrial CIES
$p_{\text{mt}}^{\text{max}}$ (kW)	1000	650	1400
$p_{\text{mt}}^{\text{min}}$ (kW)	200	100	200
$r_{\text{mt}}^d / r_{\text{mt}}^u$ (kW)	500	300	700
$p_{\text{eb}}^{\text{max}}$ (kW)	400	400	900
$r_{\text{eb}}^d / r_{\text{eb}}^u$ (kW)	200	200	400
$p_{\text{ac}}^{\text{max}}$ (kW)	500	500	800
$p_{\text{ac}}^{\text{min}}$ (kW)	60	60	80
$r_{\text{ac}}^d / r_{\text{ac}}^u$ (kW)	200	200	400
$p_{\text{ec}}^{\text{max}}$ (kW)	100	200	500
$r_{\text{ec}}^d / r_{\text{ec}}^u$ (kW)	50	100	200
$\phi_{\text{CIES}}^u / \phi_{\text{CIES}}^d$ (kW)	200	200	500
γ_{gas} (\$/m ³)	0.45	0.55	0.55
β_{CIES}	0.95	0.95	0.95

**Fig. 5.** TOU price and main grid price.

$$\begin{aligned}
 & \left| P_{n,t,k}^{\text{ADN,buy}} - P_{n,t,k}^{\text{ADN,sell}} - \left(P_{n,t,k}^{\text{CIES,buy}} - P_{n,t,k}^{\text{CIES,sell}} \right) \right| \leq \varepsilon_1 \\
 & \left| \left(C_k^{\text{ADN}} + \sum_{n=1}^N C_{n,k}^{\text{CIES}} \right) - \left(C_{k-1}^{\text{ADN}} + \sum_{n=1}^N C_{n,k-1}^{\text{CIES}} \right) \right| \leq \varepsilon_2
 \end{aligned} \quad (28)$$

where, ε_1 and ε_2 are convergence accuracies. Equation (28) indicates that in the k th iteration, the tie-line powers of the ADN and MCIE are as coupled variables and their difference should satisfy the accuracy requirement. In addition, the fluctuation of the total costs of the ADN containing MCIESs in the k th and $k-1$ st iterations should satisfy the accuracy requirement, thus ensuring the overall benefit of the ADN.

If the convergence criteria cannot be satisfied simultaneously, the penalty function multipliers are updated according to Eq. (29).

$$\begin{aligned}
 \omega_{n,t,k} &= \omega_{n,t,k-1} + \chi_{n,t,k-1} \left| P_{n,t,k}^{\text{ADN,buy}} - P_{n,t,k}^{\text{ADN,sell}} - \left(P_{n,t,k}^{\text{CIES,buy}} - P_{n,t,k}^{\text{CIES,sell}} \right) \right| \\
 \chi_{n,t,k} &= \lambda \chi_{n,t,k-1}
 \end{aligned} \quad (29)$$

where, $\chi_{n,t,k}$ is the first-order term multiplier of the k th iteration during period t ; λ is the iterative multiplier, which is generally in the range of $2 < \lambda < 3$. Since only the tie-line powers and transaction price need to be transmitted at each iteration, the privacy of each stakeholder is effectively protected.

6. Case study

The proposed expansion planning approach is tested on a 54-node distribution system, where the candidate installation nodes for WTs, PVs, MTs, and BESs are 12, 14, 17, 24, 39, and 47; 5, 8, 10, 26, 30, 33, and 37; 9, 13, 16, and 32; 8, 9, 13, 16, 17, 32, and 50, respectively. The candidate newly built lines for SOPs are 22–23, 6–28, 30–43, 33–39, 46–47, 2–8, and 10–15. The residential CIES, commercial CIES, and industrial CIES are connected to nodes 4, 40, and 20, and it is assumed that the loads in the above CIESs are represented by residential, commercial, and industrial loads, respectively. The CWGAN-GP model is built via the deep learning framework Pytorch, the measured REG output data from the Elia integration datasets from June 1, 2021 to May 31, 2023 are selected as the sample set, and the continuous data of one day are taken as one-day samples, from which 80 % of the samples are randomly selected for training and the remaining 20 % are taken as the testing samples. The learning rate and batch size in training are 0.0002 and 64, respectively. The substation parameters are shown in Ref. [37], the ADN parameters are listed in Table 2, the MCIES parameters are listed in Table 3 and [33], and the time-of-use (TOU) and main grid prices are shown in Fig. 5. The initial values of χ_n and ω_n are 0.0001 and 0.02, ε_1 and ε_2 are set to 0.01, $T = 24$ h, and $\Delta t = 1$ h. The planning procedure is performed using MATLAB R2018a with the CPLEX 12 solver.

To verify the effectiveness and superiority of the proposed model, the ADN planning with and without consideration of optimal MCIES operation is compared. In addition, SOPs, as flexible power electronic devices connecting different power supply areas of an ADN, demonstrated a significant impact on the network flexibility. Consequently, four ADN planning modes are compared in this study. Mode 1 represents the conventional planning method that does not consider MCIES optimal operation and SOP access. Mode 2 enhances M1 by incorporating MCIES optimal operation into the ADN planning. Mode 3 improves upon Mode 1 by configuring SOPs. Mode 4 i.e. the proposed model combines MCIES optimal operation and SOP access in ADN planning.

6.1. Scenario generation analysis

The effectiveness of scenario generation is analyzed in terms of seasonal characteristics, diversity and correlation, as well as typical daily scenarios of REGs.

6.1.1. Seasonal characteristics statistical analysis

To demonstrate the effectiveness of the CWGAN-GP scenario generation method in portraying the seasonal output characteristics of REGs, a comparison is made between the average daily generation of real and generated samples of the WT and PV plants across different seasons. This comparison is depicted in Fig. 6.

As shown in Fig. 6, the average daily generation of the generated samples under different seasons is similar to that of the historical samples and captured the significant seasonal difference—WTs generate more electricity in winter and less in summer, while PV generate more electricity in summer and less in winter. These results demonstrate that the proposed CWGAN-GP-based scenario generation can accurately capture the seasonal output characteristics of REGs and effectively track

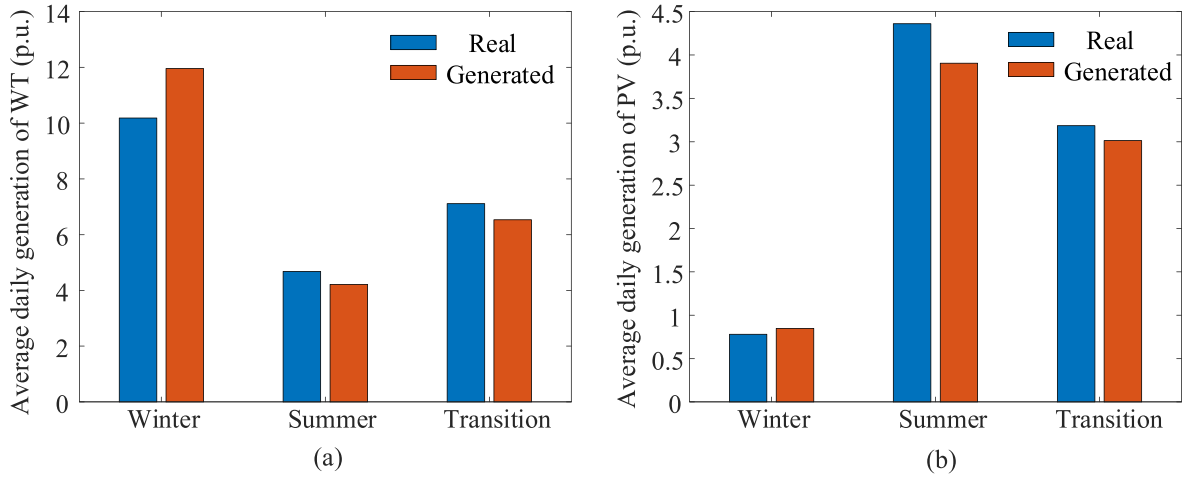


Fig. 6. Average daily power generation of (a) WT and (b) PV under different seasons.

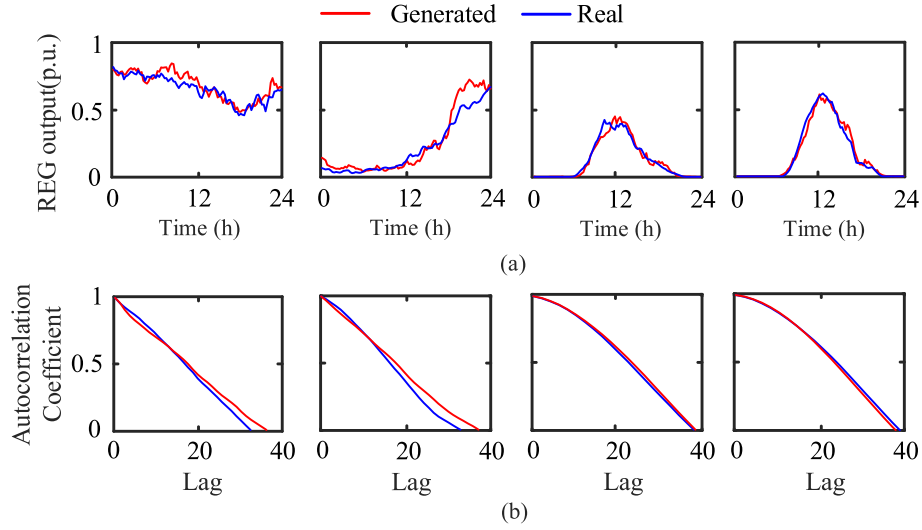


Fig. 7. Pairs of generated and real samples of REGs and their autocorrelation coefficients.

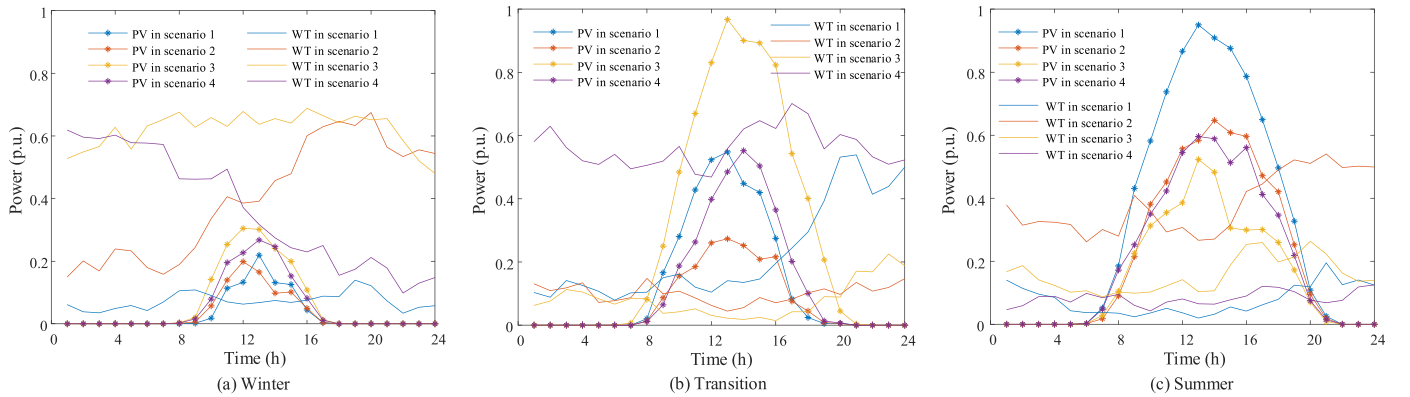


Fig. 8. Seasonal typical scenarios of REGs.

their seasonal output trends.

6.1.2. Diversity and correlation analysis

An output comparison of sample pairs of REGs selected from the generated and real samples via the Euclidean distance nearest principle and their autocorrelation coefficients are shown in Fig. 7.

Fig. 7(a) shows that the generated samples effectively capture the diversity of modal changes. However, the pairs of samples do not completely coincide, which demonstrates that the scenario generation model performs fitting instead of simply memorizing the probability distribution of the historical data. This indicates that the CWGAN-GP model is trained without overfitting, and the generated data capture

Table 4
Typical daily scenario probabilities.

Typical scenario	Scenario probability			
	Typical scenario 1	Typical scenario 2	Typical scenario 3	Typical scenario 4
Winter	0.0789	0.0493	0.0838	0.0345
Summer	0.0832	0.0555	0.0504	0.0630
Transition	0.1153	0.0953	0.1504	0.1404

the characteristics of the historical data, such as rapid changes, spikes, and daily cycle changes. Fig. 7(b) shows that the autocorrelation coefficients between the generated and real samples gradually decrease as the time interval increases and all trends are basically the same, indicating that the generated samples have temporal correlation features similar to those of the historical samples. Therefore, the generated scenarios adequately represent the actual operating environments of REGs.

6.1.3. Seasonal typical daily scenario analysis

The initial scenarios under each season generated by CWGAN-GP are reduced to four typical scenarios using K-Medoids, the seasonal typical scenarios of REGs and their probability are presented in Fig. 8 and Table 4, respectively.

From Fig. 8, it can be observed that the PV outputs are generally

lower in winter, higher in summer, and at intermediate levels in the transition season, whereas the WT outputs are the opposite and more volatile and diverse, in line with the seasonal output characteristics of REGs. This indicates that the clustering scenarios effectively reflect the seasonal and daily fluctuation characteristics of the REGs.

6.2. Expansion planning scheme analysis

The expansion planning results and schemes under the four modes are presented in Fig. 9 and Table 5, respectively.

As shown in Fig. 9 and Table 5, the proposed model realizes the coordinated expansion planning of REGs and diversified flexible resources in ADNs. Among them, two new substations are built in Modes 1 and 2 that do not consider SOPs. In contrast, as SOPs are considered in Modes 3 and 4, only one new substation is built in the ADN despite the increase in newly built lines. This is because SOPs connect the power supply areas of different substations and provide support for interconnection among different areas. Additionally, the installed capacities of MTs and BESs in Mode 4 which considers the optimal operation of MCIESs are 8398 kW and 3379 kWh, respectively. Both values are reduced compared to those in Mode 3, which are 12515 kW and 3529 kWh, respectively. The same applies to Modes 1 and 2. This is because MCIESs can be used as bi-directionally regulated flexible resources to provide flexibility support for the ADN, thus reducing the ADN

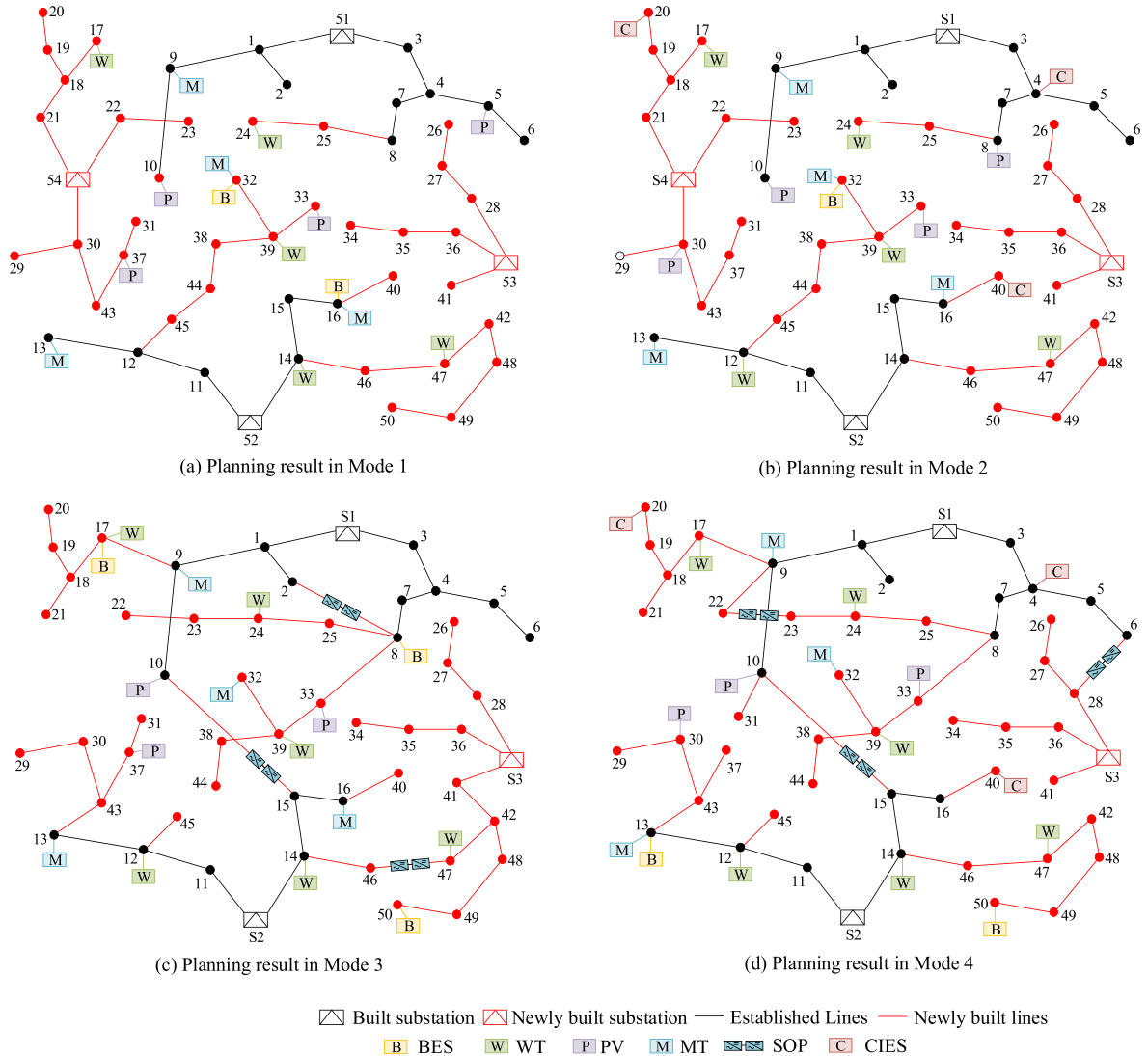


Fig. 9. ADN expansion planning results under different modes.

Table 5

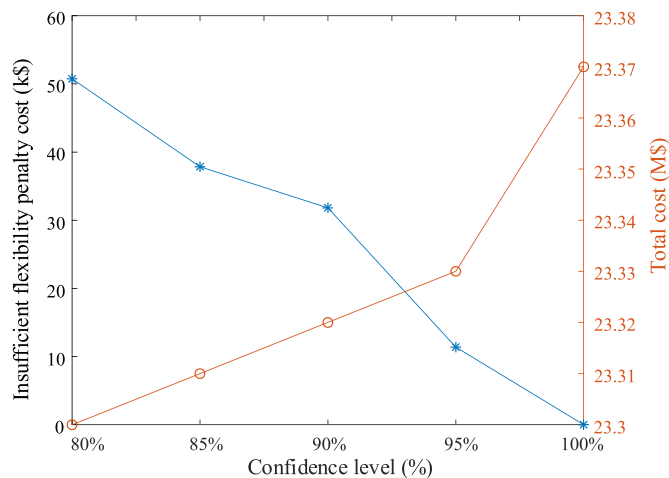
ADN expansion planning schemes under different modes.

Mode	New/expanded substations (MVA)	Newly built lines		WT/kW (bus)	PV/kW (bus)	MT/kW (bus)	BES/kWh/kW (bus)	SOP/kVA (bus)
		Same	Different					
Mode 1	S3 (22.2), S4 (22.2)	19-20, 18-19, 17-18, 18-21, 22-23, 24-25, 8-25, 33-39, 32-39, 38-39, 38-44, 26-27, 27-28, S3-28, S3-36, 35-36, 34-35, S3-41,	44-45, S4-30, S4-22, S4-21, 31-37	800(14), 5000 (17), 5000(24), 5000 (39), 3400 (47)	4700 (5), 5000 (10), 5000 (33), 1100 (37)	442 (9), 1089 (13), 1628 (16), 186 (32)	63/16 (16), 734/184 (32)	/
Mode 2	S3 (22.2), S4 (22.2)	32-39, 38-39, 38-44, 26-27, 27-28, S3-28, S3-36, 35-36, 34-35, S3-41,	44-45, S4-30, S4-22, S4-21, 31-37	200(12), 5000 (17), 5000(24), 5000 (39), 4300 (47)	4300 (8), 5000 (10), 1200 (30), 5000 (33)	442 (9), 1075 (13), 1338 (16), 301 (32)	275/69 (32)	/
Mode 3	S3 (22.2)	16-40, 14-46, 46-47, 42-47, 42-48, 48-49, 49-50, 12-45, 37-43, 30-43, 29-30	2-8, 41-42, 31-37, 9-17, 23-24, 8-33, 13-43, 10-15	4500(12), 2500 (14), 4900(17), 5000 (24), 5000(39), 5000 (47)	1300 (10), 2200 (33), 4600 (37)	3600 (9), 1690 (13), 1100 (16), 6125 (32)	1441/360 (8), 88/22 (17), 2000/500 (50)	2000 (46-47), 1400 (2-8), 1200 (10-15)
Mode 4	S3 (22.2)	49-50, 12-45, 37-43, 30-43, 29-30	9-22, 10-31 6-28, 9-17, 23-24, 8-33, 10-15, 13-43	4900(12), 5000 (14), 5000(17), 5000 (24), 5000(39), 3400 (47)	3800 (10), 900 (26), 2000 (33)	4660 (9), 669 (13), 3069 (32)	2000/500 (13), 1379/345 (50)	1600 (22-23), 2900 (6-28), 1600 (10-15)

Table 6

ADN expansion planning costs under different modes.

Annual cost ($\times 10^4$ \$)	Mode 1	Mode 2	Mode 3	Mode 4
Total cost	2505.31	2480.86	2370.72	2349.95
Total operation cost	1886.79	1862.91	1869.32	1850.22
Total investment cost	618.52	617.95	501.40	499.73
Line investment cost	18.35	18.35	21.36	22.11
Substation investment cost	348.00	348.00	174.00	174.00
SOP investment cost	0	0	6.79	9.01
PV investment cost	84.49	82.88	43.31	35.83
WT investment cost	159.18	161.67	223.02	234.63
BES investment cost	1.62	0.56	7.17	6.87
MT investment cost	6.88	6.49	25.75	17.28
Power purchase cost from the main grid	1768.83	1796.6	1492.65	1602.07
Fuel cost of MT	71.10	71.00	305.54	192.06
O&M cost	45.42	45.09	69.68	63.93
Flexible resource interaction cost with MCIES	0	-51.23	0	-8.98
Insufficient flexibility penalty cost	1.44	1.45	1.45	1.14

**Fig. 10.** Insufficient flexibility penalty cost and total cost under different confidence levels.**Table 7**

Network flexibility indicators under different modes.

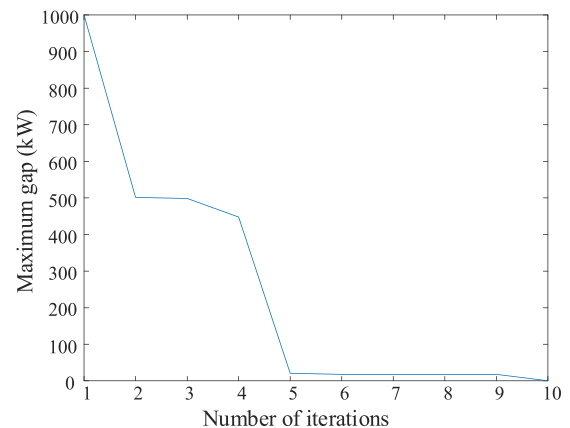
Mode	Average branch load ratio (%)	Load uniformity (%)	Average voltage deviation (%)
Mode 1	21.21	34	4.03
Mode 2	23.63	30.13	4.25
Mode 3	19.19	25.18	3.51
Mode 4	15.47	22.64	2.72

configuration for other flexible resources while meeting the load demand.

6.3. Economic analysis

The total cost and details of the ADN expansion planning under different modes are given in Table 6.

Table 6 indicates that Mode 4 (the proposed model) exhibits the lowest insufficient flexibility penalty cost and total cost, whereas Mode 1 (without consideration of the MCIES and SOPs) has the highest total

**Fig. 11.** ATC iteration curve.

cost. Compared with Mode 4, Mode 3 does not consider the optimal operation of the MCIES, resulting in a decrease in the sales revenue of the ADN to the MCIES. Additionally, due to the lack of flexibility support from the MCIES to ADN, the initial investment cost of flexible resources is increased. Two new substations are built in Mode 2 (without considering SOPs) to meet the load demand. Although this improves the interaction benefits with the MCIES, the investment cost of the substations and the power purchase cost from the main network are increased, resulting in a higher total cost. This indicates that the proposed model improves the flexibility and mutual aid ability of various power supply areas in the ADN by considering SOPs and the MCIES, while ensuring economic efficiency and improving the operational flexibility of the entire ADN.

6.4. Insufficient flexibility risk analysis

To analyze the impact of insufficient flexibility risk on ADN expansion planning, a sensitivity analysis on confidence level of the CVaR model is conducted, which is shown in Fig. 10.

The confidence level reflects the flexibility planning requirements of ADNs. Fig. 10 shows that as the confidence level increases, the insufficient flexibility penalty cost of the ADN gradually decreases, whereas the total cost gradually increases; above the 95 % confidence level, the total cost increases significantly. This is because the increase in system operational flexibility comes at the expense of the economy, highlighting the need to balance the two factors by leveraging the indicator of insufficient flexibility risk.

6.5. Network flexibility analysis

In addition to supply-demand flexibility, ADNs should also have superior network flexibility to ensure the effective transmission of power. Therefore, the average branch load ratio, load uniformity, and average voltage deviation indices of ADN are utilized to quantify and evaluate the network flexibility, which is presented in Table 7.

From Table 7, it is evident that the average branch load ratio is the lowest in Mode 4, i.e., the branch load margin is the largest, indicating that MCIESs and SOPs can improve the load distribution of branches and transfer capacity of lines. Compared with Mode 1, the load uniformity of Mode 4 decreases from 34 % to 22.64 %. The smaller the load uniformity, the more balanced the load ratio of each branch, thus strengthening the ability to cope with changes in system uncertainty. Therefore, the ADN planning scheme in Mode 4 exhibits the best network flexibility among the four modes. Moreover, the average voltage deviation of Mode 4 is 2.72 %, indicating that the proposed model effectively suppresses the voltage fluctuation and improves the ADN voltage quality compared with the other three modes.

6.6. Analytical target cascading algorithm analysis

The ATC-based model solution in this study converges after 10 iterations. Fig. 11 shows the variation in the maximum gap between the ADN and MCIES for each iteration. It can be seen that the gap between the two continuously converges to zero during the iteration process, which illustrates the effectiveness of the ATC algorithm. In addition, the calculation time of the example is 4.2 h, indicating that the algorithm is applicable for practical planning problems that require long-term decision processing. Notably, considering additional CIESs in the model will not significantly increase the calculation time owing to the parallel processing capability of the ATC method.

7. Conclusion

As coupling among diverse energy sources intensifies, various forms of CIES are emerging within ADNs. Tailoring to the attributes of MCIESs characterized by abundant flexible and controllable resources, this study

introduces a flexibility-based expansion planning approach for ADNs, which effectively utilizes the operation flexibility of MCIES to postpone ADN investment and realizes the coordinated planning of DERs and power networks, thereby improving the economy, network flexibility, and supply-demand flexibility of the overall ADN. Responding to renewable uncertainties, the developed model-free CWGAN-GP scenario generation method portrays the correlation, volatility, and uncertainty characteristics of REG outputs in different seasonal environments and does not require explicit feature model construction, which has broad applicability and provides a new approach for planning scenario generation. Finally, the proposed ATC-based distributed modeling method refines the interest game among all stakeholders regarding economic dispatch on the basis of ensuring privacy, truly reflecting the advantages of decentralized autonomy of MCIESs.

In future studies, the reactive power optimization and regulation strategy of MCIES should be further considered during ADN planning. In addition, the proposed method provides a planning strategy based on current system conditions. However, this method is widely applicable and can be adapted to the modeling of real-world systems. To realize long-term load growth or increased renewable energy integration, the proposed planning model can be re-leveraged to determine the allocation schemes for increased REGs and flexible resources.

CRedit authorship contribution statement

Chunling Wang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis. **Chunming Liu:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Xiulin Zhou:** Methodology, Investigation, Formal analysis, Conceptualization. **Gaoyuan Zhang:** Visualization, Supervision, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

References

- [1] Suryakiran BV, Nizami S, Verma A, et al. A DSO-based day-ahead market mechanism for optimal operational planning of active distribution network. *Energy* 2023;282:128902.
- [2] Huang M, Chang J, Guo A, et al. Cascade hydropower stations optimal dispatch considering flexible margin in renewable energy power system. *Energy* 2023;285:129375.
- [3] Wang J, Hu Z, Xie S. Expansion planning model of multi-energy system with the integration of active distribution network. *Appl Energy* 2019;253:113517.
- [4] Zhang D, Zhu H, Zhang H, et al. Multi-objective optimization for smart integrated energy system considering demand responses and dynamic prices. *IEEE Trans Smart Grid* 2022;13(2):1100–12.
- [5] Li Y, Feng B, Wang B, et al. Joint planning of distributed generations and energy storage in active distribution networks: a bi-level programming approach. *Energy* 2022;245:123226.
- [6] Koutsoukis N, Georgilakis P. A multistage distribution network planning method considering distributed generation active management and demand response. *IET Renew Power Gener* 2022;16(1):65–76.
- [7] Xu X, Niu D, Peng L, et al. Hierarchical multi-objective optimal planning model of active distribution network considering distributed generation and demand-side response. *Sustain Energy Technol Assessments* 2022;53:102438.
- [8] Pamshetti V, Singh S, Thakur A, et al. Cooperative operational planning model for distributed energy resources with soft open point in active distribution network. *IEEE Trans Ind Appl* 2023;59(2):2140–51.
- [9] Hemmati R. Dynamic expansion planning in active distribution grid integrated with seasonally transferred battery swapping station and solar energy. *Energy* 2023;277:127719.

- [10] Huang Z, Xu Y, Chen L, et al. Coordinated planning method considering flexible resources of active distribution network and soft open point integrated with energy storage system. *IET Gener Transm Distrib* 2023;17(23):5273–85.
- [11] Tian Z, Li X, Niu J, et al. Enhancing operation flexibility of distributed energy systems: a flexible multi-objective optimization planning method considering long-term and temporary objectives. *Energy* 2024;288:129612.
- [12] Yang Z, Han G, Yang F, et al. A distribution network planning method considering the distributed energy resource flexibility of virtual power plants. *Sustainability* 2023;15(19):14399.
- [13] Shirsat A, Tang W. Data-driven stochastic model predictive control for DC-coupled residential PV-storage systems. *IEEE Trans Energy Convers* 2021;36(2):1435–48.
- [14] Wang B, Wang X, Zhu Z, et al. Siting and sizing of energy storage for renewable generation utilization with multi-stage dispatch under uncertainty: a tri-level model and decomposition approach. *Appl Energy* 2023;344:121286.
- [15] Wang Q, Zhang X, Yi C, et al. A novel shared energy storage planning method considering the correlation of renewable uncertainties on the supply side. *IEEE Trans Sustain Energy* 2022;13(4):2051–63.
- [16] Hassaballah E, Keshta H, Abdel-Latif K, et al. A novel strategy for real-time optimal scheduling of grid-tied microgrid considering load management and uncertainties. *Energy* 2024;299:131419.
- [17] Wang B, Wang X, Zhu Z, et al. Siting and sizing of energy storage for renewable generation utilization with multi-stage dispatch under uncertainty: a tri-level model and decomposition approach. *Appl Energy* 2023;344:121286.
- [18] Yoo J, Son Y, Yoon M, et al. A wind power scenario generation method based on Copula functions and forecast errors. *Sustainability* 2023;15(23):16536.
- [19] Dong W, Chen X, Tang Q. Data-driven scenario generation of renewable energy production based on controllable generative adversarial networks with interpretability. *Appl Energy* 2022;308:118387.
- [20] Yuan R, Wang B, Sun Y, et al. Conditional style-based generative adversarial networks for renewable scenario generation. *IEEE Trans Power Syst* 2023;38(2):1281–96.
- [21] Wang C, Liu C, Chen J, et al. Cooperative planning of renewable energy generation and multi-timescale flexible resources in active distribution networks. *Appl Energy* 2024;356:122429.
- [22] Li Y, Li J, Wang Y. Privacy-preserving spatiotemporal scenario generation of renewable energies: a federated deep generative learning approach. *IEEE Trans Ind Inf* 2022;18(4):2310–20.
- [23] Chen J, Yang Y, Tang X, et al. Generative adversarial networks: a literature review 2020;14(12):4625–47.
- [24] Pan Z, Yu W, Yi X, et al. Recent progress on generative adversarial networks (GANs): a survey. *IEEE Access* 2019;7:36322–33.
- [25] Wang Z, Wang S, Zhou C, et al. AVO inversion based on closed-loop multitask conditional Wasserstein generative adversarial network. *IEEE Trans Geosci Rem Sens* 2023;61:1–13.
- [26] Gulrajani I, Ahmed F, Arjovsky M, et al. Improved training of Wasserstein GANs. *NIPS*; 2017. p. 30.
- [27] Badoozadeh S, Nikdel N, Galvani S, et al. Probabilistic optimal power flow in wind energy integrated power system based on the K-medoids data clustering method considering correlated uncertain variables. *IET Renew Power Gener* 2023;17(13):3179–94.
- [28] Zhang J, Foley A, Wang S. Optimal planning of a soft open point in a distribution network subject to typhoons. *Int J Electr Power Energy Syst* 2021;129:106839.
- [29] Ehsanbakhsh M, Sepasian M. Simultaneous siting and sizing of soft open points and the allocation of tie switches in active distribution network considering network reconfiguration 2022;17(1):263–80.
- [30] Fan W, Tan Z, Li F, et al. A two-stage optimal scheduling model of integrated energy system based on CVaR theory implementing integrated demand response. *Energy* 2023;263:125783.
- [31] Long C, Wu J, Thomas L, et al. Optimal operation of soft open points in medium voltage electrical distribution networks with distributed generation. *Appl Energy* 2016;184:427–37.
- [32] Cao W, Wu J, Jenkins N, et al. Operating principle of soft open points for electrical distribution network operation. *Appl Energy* 2016;164:245–57.
- [33] Liu C, Wang C, Yin Y, et al. Bi-level dispatch and control strategy based on model predictive control for community integrated energy system considering dynamic response performance. *Appl Energy* 2022;310:118641.
- [34] Li Y, Wang B, Yang Z, et al. Optimal scheduling of integrated demand response-enabled community-integrated energy systems in uncertain environments. *IEEE Trans Ind Appl* 2022;58(2):2640–51.
- [35] Li Y, Wang C, Li G, et al. Improving operational flexibility of integrated energy system with uncertain renewable generations considering thermal inertia of buildings. *Energy Convers Manag* 2020;207:112526.
- [36] Qiu H, Gu W, Xu Y, et al. Robustly multi-microgrid scheduling: stakeholder-parallelizing distributed optimization. *IEEE Trans Sustain Energy* 2020;11(2):988–1001.
- [37] Zhang S, Wang H, Li R, et al. Active distribution network expansion planning method considering the integration of soft open point. *Proceedings of the CSEE* 2023;43(1):48–60 (in Chinese).